GOVERNMENT OF KARNATAKA

Department of Collegiate And Technical Education

GOVERNMENT POLYTECHNIC FOR WOMEN, BENGALURU

Department of Computer Science and Engineering

CIE PHASE - 3

Submitted by

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6th Sem Computer Science and Engineering

Cohort Owner

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Department of Computer Science & Engineering

GWPT, BENGALURU

2025–2026

GOVERNMENT POLYTECHNIC FOR WOMEN, BENGALURU

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that **Ms. APOORVA C (119CS23005),** student of sixth semester Computer Science and Engineering, Government Polytechnic for Women, Bengaluru has partially fulfilled the requirements for the Diploma in Computer Science and Engineering award for the year 2025–2026, by submitting a report of CIE Phase 3.

Cohort Owner

Mrs. Rekha A Naik, Selection Grade Lecturer

Department of Computer Science and Engineering

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1. Select any one job role of his/her interest in an organization or role assigned by the training supervisor:

Role: AI Application Developer (Healthcare System)

An AI Application Developer is responsible for designing and developing intelligent applications by integrating Artificial Intelligence with web technologies. The role involves building user-friendly systems, integrating APIs, and deploying AI-based solutions for real-world problems.

In this project, the intern worked on a **Vitamin Deficiency Detection System using AI and Flask**, where the system analyzes user symptoms and provides health suggestions using an intelligent application.

Responsibilities:

Application Development

- Designing a web-based interface using HTML and CSS
- Developing the backend using the Flask framework
- Creating a user-friendly system for symptom input

API Integration

- Integrating AI-based API (DeepSeek) for prediction
- Handling API requests and responses
- Managing JSON data for communication

System Development

- Implementing a symptom-based prediction system
- Developing logic for vitamin deficiency detection
- Creating multiple interaction features

Data Handling

- Storing user predictions in JSON files

- Managing feedback data
- Handling file uploads and storage

Testing & Debugging

- Testing the system with different inputs
- Handling API errors using a fallback system
- Fixing bugs and improving performance

Continuous Learning

- Learning Flask and API integration
- Understanding real-time AI applications
- Improving system features and usability

1.1 Intern's Ability to Apply the Skill and Technical Knowledge in OJT-2

During my OJT-2 internship, I worked on developing an AI-based Vitamin Deficiency Detection System using Flask and API integration. The main objective of the project was to create a smart web application that analyzes user symptoms and provides suggestions related to vitamin deficiencies. This system helps users understand possible health issues and receive basic guidance through a digital platform.

During this internship, I applied my knowledge of Python programming and the Flask framework to develop the backend of the application. Flask was used to create routes that connect the user interface with the backend system. I implemented routes such as **/predict** and **/api/predict** to process user inputs and generate responses. This helped me understand how web applications handle requests and return results.

I also learned how to integrate external AI services using APIs. The system sends the user symptoms to an AI model which analyzes the data and returns structured information. The response includes details such as vitamin name, severity, food suggestions, and precautions in JSON format. This improved my understanding of API communication and real-time data processing.

Additionally, I implemented data storage using JSON files to store prediction history and user feedback. This helped in managing application data efficiently and maintaining records of user interactions. I also improved the system by adding multiple interaction features to enhance the user experience...

Overall, this internship helped me strengthen my skills in AI application development, backend programming, API integration, and system design. It provided practical experience in developing a real-world healthcare support application

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1.1 Performance on Assigned Tasks and Project

During my OJT-2 internship, I worked on developing an AI-based **Vitamin Deficiency Detection System** using Python, Flask, and API integration. The main objective of the project was to create a smart web application that analyzes user symptoms and provides suggestions related to vitamin deficiencies. This system helps users understand possible health issues and receive basic guidance through a digital platform

During the internship, I applied my knowledge of **Python programming and the Flask framework** to develop the backend of the application. Flask was used to create routes that connect the user interface with the backend system. I implemented routes such as **/predict** and **/api/predict** to process user inputs and generate responses. This helped me understand how web applications handle requests and return results

I also learned how to integrate external **AI services using APIs**. The system sends the user symptoms to an AI model which analyzes the data and returns structured information. The response includes details such as **vitamin name, severity, food suggestions, and precautions** in JSON format This improved my understanding of API communication and real-time data processing.

Additionally, I implemented **data storage using JSON files** to store prediction history and user feedback. This helped in managing application data efficiently and maintaining records of user interactions. I also improved the system by adding **chatbot interaction features**, which guide users in entering symptoms and getting prediction results easily.

I also implemented error-handling techniques to improve system reliability. When the API fails or encounters errors, the system uses a fallback logic to generate responses. This ensures that the application continues to function without interruption and provides a better user experience.

Overall, this internship helped me strengthen my skills in **AI application development, backend programming, API integration, and system design**. It provided practical experience in developing a real-world healthcare support application.

1.2 Ability to Add Value to the Organization Through Internship

The developed **Vitamin Deficiency Detection System** adds significant value to the organization by providing an intelligent and automated solution for basic health assessment. In traditional methods, identifying vitamin deficiencies requires clinical diagnosis, laboratory tests, and expert consultation, which can be time-consuming and costly. This system simplifies the initial stage of health analysis by allowing users to enter their symptoms and receive instant suggestions, thereby reducing time and manual effort

The integration of **Artificial Intelligence (AI)** improves the efficiency of the system. By analyzing user symptoms, the application can provide structured outputs such as possible vitamin deficiency, severity level, recommended foods, and necessary precautions This helps users quickly understand their health condition and encourages early awareness and preventive care.

Another important value addition is the **user-friendly interface** of the system. Users can easily enter symptoms and receive results without technical difficulty. The system also includes **chatbot interaction features**, which guide users during symptom entry and make the application more interactive and easy to use.

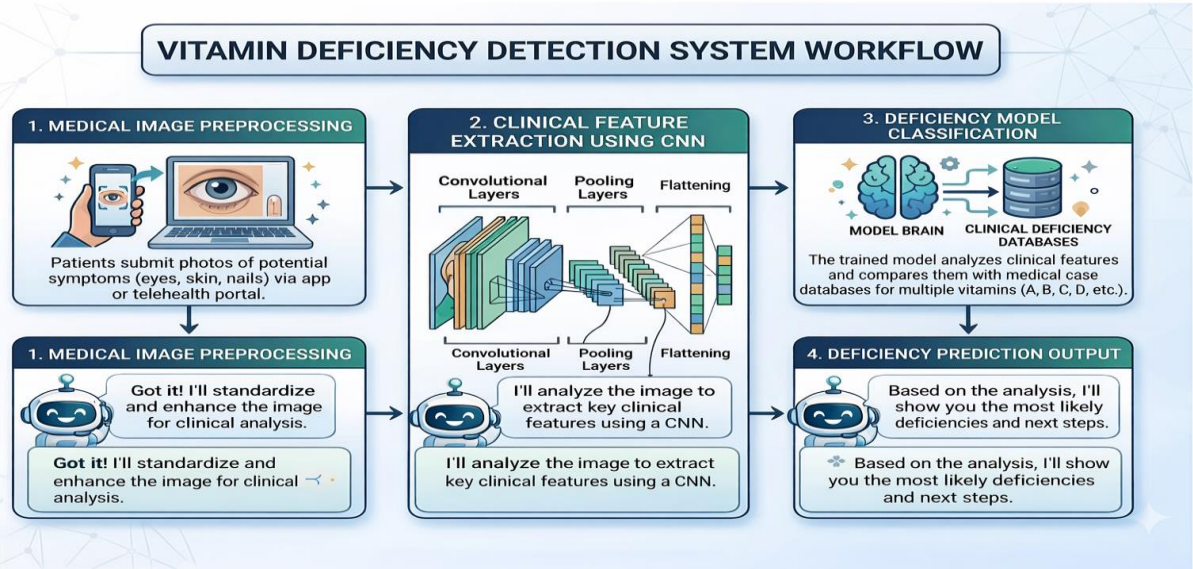
The project also demonstrates the practical implementation of modern technologies such as **web development and AI integration**. By combining a Flask-based backend with API-based intelligence, the system represents a real-world application of technology in the healthcare domain. This supports the organization's initiative toward **digital transformation and smart systems**.

In addition, the system stores **prediction history and user feedback using JSON files**, which helps in better data management and future analysis. Error handling mechanisms are also implemented to ensure the system continues to function smoothly even if external services fail, improving reliability and robustness.

Overall, this internship project contributes to the organization by **improving efficiency, enhancing user experience, , and demonstrating the potential of AI-driven healthcare solutions**. It highlights how technology can be used effectively to solve real-world problems and supports the development of innovative intelligent systems.

2. Document a Use case on a task where he is working as intern

2.1 Use case:



The system allows users to enter their symptoms through a simple and user-friendly web interface. The interface is designed to be easy to use so that users can clearly describe their health issues without any confusion. This makes the system accessible to different types of users, including those with basic technical knowledge.

Once the symptoms are submitted, the backend developed using **Flask** processes the input data. The system first validates the entered symptoms and ensures that the input is meaningful. After validation, the data is. sent to an **AI-based API** for further analysis.

he AI-based API analyzes the symptoms and returns a structured response in **JSON format**. The response includes details such as predicted vitamin deficiency, severity level, recommended foods, precautions, and confidence score. This structured format helps organize the information clearly

Finally, the system displays the result to the user in a readable format. The user can easily understand their possible health condition and follow the suggested recommendations

This use case demonstrates how the system provides **quick, accurate, and helpful health suggestions using AI technology**.

2.2 Approach:

1. Input Collection

The user enters symptoms through the web interface. The interface is designed to be simple and user-friendly so that users can easily describe their health issues without confusion.

2. Data Processing

The input data is processed and validated using the Flask backend. The system checks for empty or incorrect inputs before sending the data for analysis, ensuring that only meaningful data is processed.

3. API Integration:

The symptoms are sent to an AI-based API for analysis. The backend communicates with the API using a structured request format, enabling intelligent symptom analysis using AI models..

.**4. Response Handling**

The API returns a response in JSON format containing prediction details. The system extracts important information such as vitamin name, severity level, and recommendations..

5. Output Display

The result is displayed to the user with predicted vitamin deficiency and health suggestions. The output also includes recommended foods and precautions for better health awareness.

6. Data Storage

User inputs and prediction results are stored in JSON files. The system maintains prediction history for future reference and better data management..

7. Error Handling

A fallback system is implemented when the API fails. The system detects API errors and switches to mock analysis to ensure continuous system operation..

8. Deployment

The system is deployed as a web application using Flask. Users can access the application through a web browser, making the system practical and easy to use.